# Dmytro Barannik, PhD

**Staff Software Engineer & Tech Lead**

📍 Ukraine (UTC+2 / EET) · Worldwide Remote (Full-time / B2B Contractor) | ✉️ d.v.barannik@gmail.com | 📱 +380 66 174 82 28 |
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## EXECUTIVE PROFILE

**Staff Software Engineer, Tech Lead & Systems Architect** with **9+ years of engineering leadership** spanning high-load streaming media, autonomous AI swarm orchestration, zero-allocation C++20 systems, multiplatform mobile & edge web, and applied cryptographic compression research. Proven track record scaling client architectures for millions of active users at **MEGOGO**, leading enterprise teams across Smart Home & Healthcare at **CHI Software**, bootstrapping indie software ventures to **25,000+ downloads with a 5.0★ rating** (**DiMALiNA**), and inventing state-patented algebraic compression algorithms. Expert at directing multi-agent AI systems, conducting high-level technical reviews, and eliminating architectural debt.

**CORE TECH STACK:** **Autonomous AI Agent Swarms**, Kotlin & KMP, Modern C++20 (SIMD AVX2/NEON), High-Load Streaming (ExoPlayer), System Architecture, Astro & Next.js, IoT (ESP32/LoRa), Applied InfoSec & Cryptography.

## COMMERCIAL EXPERIENCE & TECHNICAL LEADERSHIP

### Founder & Principal Software Engineer — DiMALiNA Studio — Commercial Indie — 09.2017 – Present

- Bootstrapped and shipped independent commercial software products with **25,000+ total downloads** and a perfect **5.0★ rating** on Google Play (*Tic-Tac-Toe: 3 Player Game*, *Brainrot Clicker*, *People Notes AI CRM*).
- Owned entire end-to-end engineering lifecycles solo: UX/UI design in Figma, native Jetpack Compose & Clean Architecture, offline-first Room databases, and multi-tier Coroutine flows.
- Architected real-time multiplayer backend pipelines using **Firebase Firestore, Cloud Functions**, and seamless offline data synchronization.
- Engineered commercial monetization models: integrated Google Play Billing (IAP subscriptions/consumables) and optimized AdMob mediation waterfalls.
- Orchestrated autonomous AI agent swarms and multi-agent development architectures (spec-driven synthesis, automated QA, zero regressions), accelerating delivery velocity by **3x** from conception to production.

### Senior Software Engineer (Streaming & Android) — MEGOGO — High-Load OTT Media — 03.2025 – 11.2025

- Engineered high-load client streaming pipelines for a premier OTT streaming platform serving **millions of active users** across mobile and TV ecosystems.
- Migrated legacy XML-based screens to modern declarative **Jetpack Compose**, achieving a **~35% boost in UI rendering performance** and eliminating UI jank during rapid catalog scrolling.
- Implemented channel availability, parental locks, and playback entry workflows utilizing **ExoPlayer** with custom low-latency adaptive buffering.
- Maintained **99.8% crash-free sessions** across millions of active streams through rigorous lifecycle scoping, profiler memory leak audits, and automated regression testing.
- Engineered complex deep-linking attribution capabilities via Install Referrer API and customized URI scheme dispatchers, maximizing acquisition funnels.

### Lead Software Engineer / Tech Lead — CHI Software — Enterprise Consultancy — 09.2020 – 03.2025

- Spearheaded distributed engineering teams delivering mission-critical applications across Healthcare, Smart Home, Security, and Media domains.
- Integrated high-performance C++ ALPR (Automatic License Plate Recognition) OCR engine into Android via JNI/NDK, optimizing Camera 2 image streams in real time.
- Architected the remote configuration platform **LiftApp** utilizing Wi-Fi Direct and BLE, enabling seamless zero-config peer-to-peer hardware control.
- Orchestrated architectural migrations toward reactive unidirectional data flows (MVI/MVVM) and multi-module structures, reducing incremental build times by **~40%**.
- Mentored 10+ junior and mid developers, authored architectural guidelines, and instituted automated CI/CD quality gates, drastically reducing bug density.

### Senior Software Engineer (Mobile & Systems) — Nitrix Studio — Commercial Agency — 08.2018 – 08.2020

- Engineered cross-platform client solutions (Flutter BLoC & Native Android) and backend microservices with real-time WebSocket messaging and SQLite/Room caching.
- Designed scalable domain architectures, authored comprehensive automated testing suites, and standardized deployment automations.

### Software Engineer — IT Company — Commercial Software — 09.2017 – 06.2018

- Developed native Android client applications in Java and Kotlin following Material Design guidelines and reactive event patterns.
- Integrated third-party RESTful APIs, location services, and background synchronization workers.

## TECHNICAL STACK & ARCHITECTURAL EXPERTISE

**AUTONOMOUS AI & ARCHITECTURE**
**AI Agent Swarms & Orchestration**, Spec-Driven Synthesis, Clean / Hexagonal / Event-Driven Systems, Decomposition, Zero-Regression Gates.

**MOBILE & MULTIPLATFORM**
**Kotlin (Expert), Jetpack Compose, Material 3**, Kotlin Multiplatform (KMP), Flutter (Dart), Coroutines, Flow, Room, ExoPlayer, Android NDK/JNI.

**SYSTEMS & LOW-LEVEL C++**
**Modern C++20/23, SIMD (AVX2/NEON)**, Zero-Allocation Memory Buffers, Memory-Mapped I/O, POSIX/Linux, CMake, Cross-Compilation Toolchains.

**WEB, CLOUD & REAL-TIME**
**Astro Islands, Next.js, Node.js**, WebSockets, Firebase Firestore & Cloud Functions, RESTful APIs, SQLite, Sub-Second LCP Web Optimization.

**IOT & EMBEDDED TELEMETRY**
**ESP32, LoRaWAN / LoRa Mesh**, BLE (Bluetooth Low Energy), Wi-Fi Direct, MQTT, Microcontroller Firmware, Hardware Sensor Calibration.

**INFOSEC & CRYPTOGRAPHY**
**Algebraic Polynomial Lossless Compression**, Steganographic Detection, Cryptographic Protocols, Digital Forensics, Covert Channel Prevention.

## ACADEMIC LEADERSHIP & SCIENTIFIC RESEARCH

### Associate Professor (KhNUIA) · Former Lecturer (MITI) KhNUIA & MITI Academic & R&D 08.2025 – Present

Lecturing and designing advanced curricula in cybersecurity, digital forensics, and communication networks at KhNUIA (2026–Present) and MITI (2025–2026). Directing research in covert channel detection and automated incident response.

**State Patent UA No. 144599: Non-Equilibrium Video Data Compression** | Ukrainian State Patent Office
Co-inventor of patented positional video coding algorithm reducing stream information intensity; forms algorithmic core of the LibPolyCodec compression engine with SIMD AVX2 acceleration and zero memory allocation.

**Cabinet of Ministers Scholarship for Young Scientists & IEEE Committee Member** | National Honor
Awarded prestigious Cabinet of Ministers of Ukraine Scholarship for Young Scientists (2025). Technical Program Committee (TPC) Member and reviewer for the IEEE 5th ATIT 2024 International Conference.

**Scientific Publications & Monograph** | Scopus, IEEE, Academic Press
Author of **1 fundamental academic monograph** (*Fundamentals of the theory of structural-combinatorial steganographic coding*, 2017) and **20+ peer-reviewed scientific papers** in data hiding and compression.

## ARCHITECTURAL & OPEN-SOURCE HIGHLIGHTS

**LibPolyCodec: Data Compression Library ↗** | C++ Software Architect · C++20, SIMD (AVX2/NEON), Zero-Allocation, MITI
High-performance compression library utilizing algebraic polynomial coefficients, achieving sub-millisecond throughput with zero dynamic allocations in critical inner loops.

**Distributed IoT LoRa Sensor Network ↗** | IoT Firmware Engineer · C++, ESP32, LoRa (SX1276), MQTT, AES-128
Designed resilient multi-hop off-grid emergency telemetry mesh network featuring dynamic routing, AES-128 encryption, and ultra-low power consumption for field sensor arrays.

**ESMEE — Bionic Prosthetics Platform ↗** | Lead Web Architect · Next.js, React, Cloudflare Pages, WCAG 2.1 AA
Engineered modular medical portal and patient intake ecosystem for powered upper-limb prosthetics with edge deployment on Cloudflare Pages, sub-second LCP, and medical schema markup.

**ProKlimat — HVAC Platform ↗** | Lead Frontend Engineer · Astro Islands, Cloudflare Pages, 100 Lighthouse
Architected enterprise commercial HVAC portal using Astro Islands, delivering zero-layout-shift kinetic UI, persistent dark/light theming, and perfect 100 Google Lighthouse audit scores.

## EDUCATION & CREDENTIALS

**Doctor of Philosophy (PhD) in Information Security** · Structural Steganographic Polyadic Coding — 2021 – 2025
Kharkiv National University of Radio Electronics (NURE) / KhNUIA

**Master of Science (MS) in Cybersecurity (With Distinction)** — 2019 – 2021
Kharkiv National University of Internal Affairs (KhNUIA)

**Bachelor of Science (BS) in Cybersecurity** — 2015 – 2019
Kharkiv National University of Internal Affairs (KhNUIA)

## LANGUAGES & ACADEMIC PROFILES

**Languages:** **English** (Professional Working Proficiency / C1 · Full Technical & Executive Communication), **Ukrainian** (Native).
**Academic Profiles:** Scopus Author ID: 57193610360 · ORCID: 0000-0002-7074-9864 · Google Scholar Profile.